Analysis of IMF Export Trade Data Structure with WSBM

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Abstract

This study searches for structure between a global trade network of countries from the magnitude and existence of their exports. The approach leverages a Weighted and Degree Corrected Stochastic Block Model (DC-WSBM). By using export amounts as edge weights, the WSBM groups countries into fitted "blocks" based on the aspects of their export patterns. We find from this technique a latent structure of the network. We performed this approach on the International Monetary Fund export data, focusing on the latest 2023 data. The dataset featured 210 countries (nodes) and 25,694 export pairings (edges) between them. The Flat Blockmodel, with a Exponential covariate type was found to find the best fit according to our measure of error, minimum description length (MDL).

1 Introduction

The International Monetary Fund (IMF) export dataset provides a high level view of the global trade network. Its data is reflective of consolidated trade into one export amount from country or grouping like (Asia, Europe) to another. This export data is continuous, and this has an implication on how it should be approached. The typical SBM approach to modeling a network considers the unweighted connections, the existence or absence of connections, having or not having an edge between nodes. This fails to consider the varying magnitude of the weighted connections. A way to model this continuous data with a typical SBM would be to threshold these magnitudes and consider the existence or absence of connections when thresholded. This can lose a significant amount of the continuous information between the thresholds. When focusing on these connections rather than magnitudes a low export partner can seem as significant as a very high amount export partner.

Weighted Stochastic Block Models (WSBM) [1, 2] specifically address this magnitude-versus-existence issue. By considering edge weights (export amounts) into the model fitting, WSBMs can partition countries into groups that follow these patterns in the continuous export amounts. This allows the model to fit structures showing each export partner and the export amounts. This provides a better approximation of the network compared to the typical SBM.

Conveniently, the graph-tool [5] functions provide a non-parametric fitting method. This allows us to fit this WSBM model without needing to provide the function parameters to guide the fitting process.

We apply the nonparametric Bayesian WSBM [3, 4] to the 2023 IMF country-to-country export network. We want to know what latent block structure underlies this magnitude of exports between countries. We compare two different models of structure, Flat versus nested (hierarchical), and two different assumed distributions for our edge weights (Exponential versus Log-Normal). To judge our fit, we consider the minimum descriptive length provided by the fitting function as our error. This minimum descriptive length is a tradeoff between how well the model fits the data while considering the complexity of that model. A smaller descriptive length means a better fit with a simpler model. We use this error to guide the fitness of our models and determine which model best reflects the latent structure in our data.

2 Methods

2.1 Data Representation

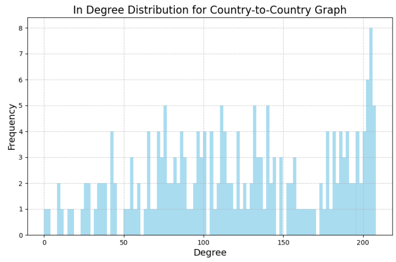
We started with the IMF export dataset. This dataset included countries and larger groupings of those countries aggregated together. To focus only on country to country, we filtered out the data associated with the larger groups, (Europe, Asia, etc...). We also filtered to only use data from one year 2023. The resulting network consisted of 210 country nodes with 25,694 directed, weighted, edges. Country A exporting X amount to Country B and Country B may or may not export a separate amount to Country A. The weights represent export amounts equivalent to millions of US dollars.

2.2 Network Characteristics

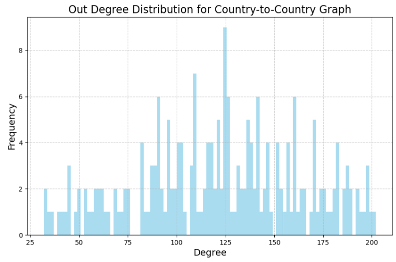
We explored the network structure, starting with the nodes (countries) and each node's degrees and weighted degrees. The unweighted node degree distributions (Figure 1) show heavy tails for in-degree (number of import partners, Figure 1a), out-degree (number of export partners, Figure 1b), and total degree (Figure 1c). This shows that most countries trade with a limited number of partners, while a few countries act as central hubs, trading with a large number of other countries.

The distribution of edge weights (export values, Figure 2) has a large variance (Figure 2a), showing that most trade relationships involve relatively small values, while a select few involve very large amounts. This distribution seems to coincide well with an Exponential Distribution. The log-transformed distribution (Figure 2b) is a centered bell shape, although slightly skewed to the right, which seems to correlate with a normal distribution, and approximately log-normal distribution. We will later test which approximation helps us configure the best fitting model.

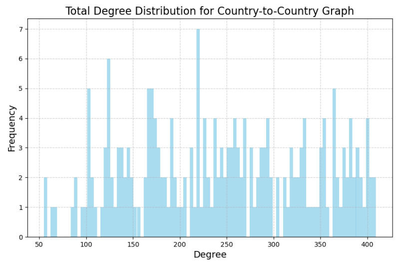
The weighted degree distributions (Figure 3), which account for the total value of exports or imports, also show pronounced long tails (Figures 3a, 3b, 3c). This suggests there are a few traders who trade a lot while most countries have small trade amounts. Looking at the whole network for context, the network density is 0.585 (Table 1), meaning that



(a) In-degree.

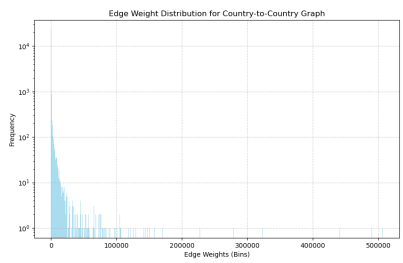


(b) Out-degree.

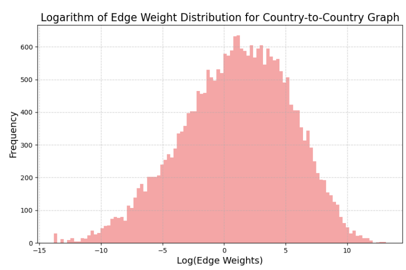


(c) Total degree.

Figure 1: Unweighted degree distributions.

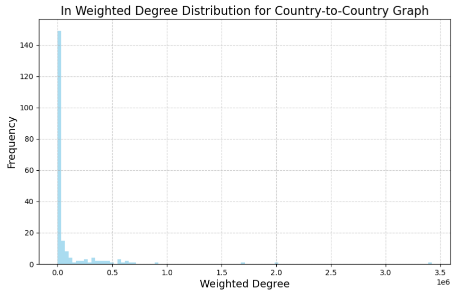


(a) Raw edge weights.

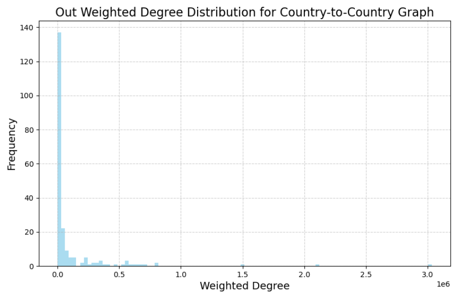


(b) Log-transformed edge weights.

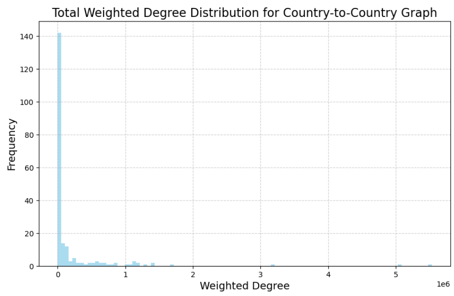
Figure 2: Edge weight distributions.



(a) Weighted In-degree.



(b) Weighted Out-degree.



(c) Weighted Total degree.

Figure 3: Weighted degree distributions.

Table 1: Global Network Metrics for the Country-to-Country Export Network

Metric	Value
Density	0.585 418
Global Clustering Coef.	3.935 935
Global Unweighted Clustering Coef.	0.825 489
Reciprocity	0.786 643
Weighted Reciprocity	0.725 315
Degree Assortativity	−0.005 949
Weighted Assortativity	−0.008 567

about 58.5% of all possible directed country-to-country trade links exist in the dataset. The reciprocity is 0.787 (Table 1), indicating a high probability that if country A exports to country B, country B also exports to country. These metrics suggest a relatively dense and interconnected global trade network but highlights that not everyone trades with each other.

2.3 WSBM Modeling

We employed the nonparametric Bayesian WSBM implementation [3, 4] available in the 'graph-tool' [5] Python library, which allows for inference of block structures without specifying many parameters beforehand. Four primary model configurations were fitted and compared:

- **Flat vs. Nested:** Flat models assume a single layer of non-overlapping communities, partitioning all nodes into distinct groups. On the other hand, nested models allow for hierarchical structures, where blocks can be partitioned into sub-blocks of a higher level block. This can potentially capture structures within other structures, such as regional groups within larger continental or economic groups.

- **Exponential vs. Log-Normal:** These covariate types specify the estimated probability distribution for the edge weights of connecting nodes. Both Exponential and Log-Normal distribution covariates are suitable for approximating continuous distributions like our export data.

The model fitting was performed using Markov Chain Monte Carlo (MCMC) algorithms implemented by 'graph-tool' [5]. These algorithms iteratively adjust block assignments to find a partitioned graph that better fits the weighted network structure. The loss function guiding the MCMC process aims to minimize the description length (MDL) [3]. MDL considers how well the partitions fit against model complexity. The model with the lowest MDL is considered the best statistical fit of the unseen structure in the network data, producing a good fit with lesser complexity.

3 Results

3.1 Model Selection

The MDL values for the four tested models were: The **Flat Exponential** model achieved

Table 2: SBM Model Fit Comparison (Minimum Description Length)

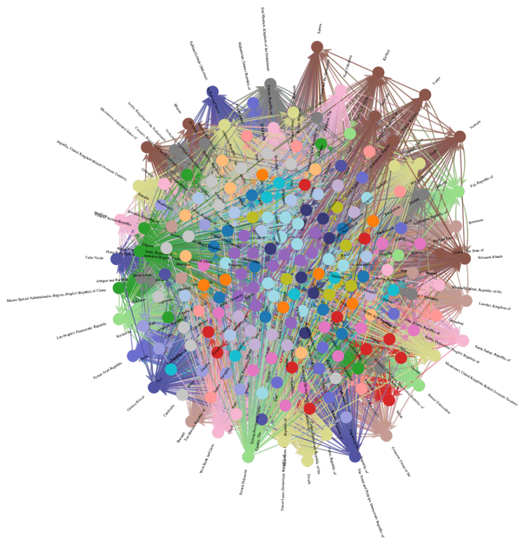
Model Configuration	MDL
Flat Exponential	71 160.8858
Flat Log-Normal	77 913.4790
Nested Exponential	72 408.1542
Nested Log-Normal	79 172.5098

the lowest Minimum Description Length (MDL = 71160.8858, Table 2), suggesting it provides the most efficient fit to the data among the tested configurations.

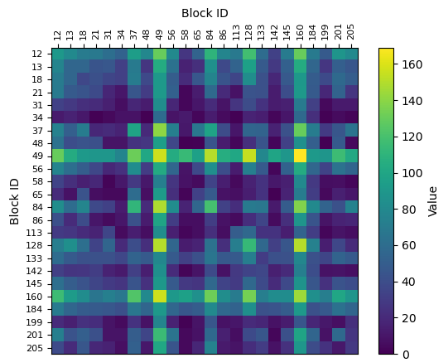
3.2 Block Memberships (Flat Exponential Model)

The Flat Exponential model divided 210 countries into 206 blocks (Table 3). Analysis of the block summary metrics confirms that 24 of these blocks were non-empty (Table 3). Figure 4 shows the visualization of this flat community structure along with the group export heatmap. The block sizes for these blocks ranged from 6 (Block 133) to 13 countries (Blocks 49 and 160), with an average size of 8.75 countries (StdDev 2.11, Table 3).

The model seemed to group countries with similar export patterns, showing differentiated economic clusters. For instance, Block 49 clearly represents a group of major global economies, including the United States, China (PRC), Germany, France, the UK, India, Italy, Spain, etc.... The commonality between them having a large amount of export trade and a large amount of trade partners. Block 160 identifies another significant cluster, primarily including Asia-Pacific economies and trade partners like Japan, South Korea, Australia, Canada, Mexico, Brazil, Indonesia, Malaysia, Singapore, Thailand, Vietnam. The specific country memberships for all 24 non-singleton blocks are found in Appendix A.



(a) Flat Exponential SBM structure.



(b) Flat Exponential parameter heatmap.

Figure 4: Visualizations for the best-fitting Flat Exponential model.

Table 3: Summary Statistics for Non-Empty Blocks Across Flat Models

Model	Total Blocks	Non-Empty Blocks	Avg Size	StdDev Size
Flat Exp.	206	24	8.75	2.11
Flat Log.	192	20	10.50	4.16

Note: Statistics calculated only for blocks containing more than one node.

Table 4: Density & Assortativity Statistics for Non-Empty Blocks in Flat Models

Model	Avg Density	StdDev Density	Avg Assortativity	StdDev Assortativity
Flat Exp.	0.6995	0.3231	-0.1741	0.1350
Flat Log.	0.7146	0.3423	-0.1297	0.1884

Note: Statistics calculated only for blocks containing more than one node. Average Local Clustering and Reciprocity omitted for brevity (Reciprocity was NaN for all).

3.3 Block Structure (Flat Exponential Model)

Table 5: Top & Bottom Inter-Block Export Flows (Flat Exponential Model)

Exporter	Importer	Ttl. Wght.	Edges
Top 5 by Total Export Value			
Block 49	Block 49	1549.42	156.00
Block 160	Block 49	1432.52	169.00
Block 49	Block 160	1337.18	156.00
Block 160	Block 160	1255.01	144.00
Block 12	Block 49	880.83	130.00
Bottom 5 by Total Export Value (Non-Zero)			
Block 199	Block 65	0.01	4.00
Block 86	Block 142	0.01	8.00
Block 142	Block 86	0.01	5.00
Block 205	Block 48	0.00	4.00
Block 21	Block 34	0.00	1.00

Total Weight represents aggregate export value (millions USD).

The heatmap of the trade between blocks of the Flat Exponential model (Figure 4b) shows the amounts that the blocks will export to each other and within themselves. Brighter yellow cells indicate greater expected export amounts while darker bluer cells indicate lesser export amounts. Looking at Block 49 (Major Global) and Block 160 (Asia-Pacific, Brazil, Canada, Mexico) we see these blocks as outstanding exporters to the other blocks as evidenced by the lighter horizontal and vertical stripes. On the diagonal we can inspect the intra-block exports, how many members of the block export to each other in the block. Block 49 (Major Global) and Block 160 (Asia-Pacific, Brazil, Canada, Mexico) show a strong yellow when trading with themselves as expected. The strongest inter-block flow is within Block 49 itself, with an aggregate weight of 1549.42 million USD (Table 5). We also see strong intra-block exports for Block 84 (Major African Nations and DPRK) and Block 128 (Caribbean, Central America, and Cambodia). These blocks are also strong traders with Block 49 (Major Global) and Block 160 (Asia-Pacific, Brazil, Canada, Mexico).

3.4 Global Network Metrics & Centrality

We also analyzed the entire network itself for baseline metrics (Table 1). Degree assortativity was found to be slightly negative (-0.006), suggesting that countries do not strongly tend to trade with other countries of similar overall connectivity (degree).

We also ran node centrality metrics on the partitioned network looking for the central hubs in each block. In Block 49 (Major Global) PageRank found Germany (0.0842, Rank 1), China (0.0833, Rank 2), and the United States (0.0819, Rank 3). In Block 160 (South Asia and Mexico and Canada) Japan (0.0908, Rank 1), the Republic of Korea (0.0892, Rank 2), and Singapore (0.0880, Rank 3).

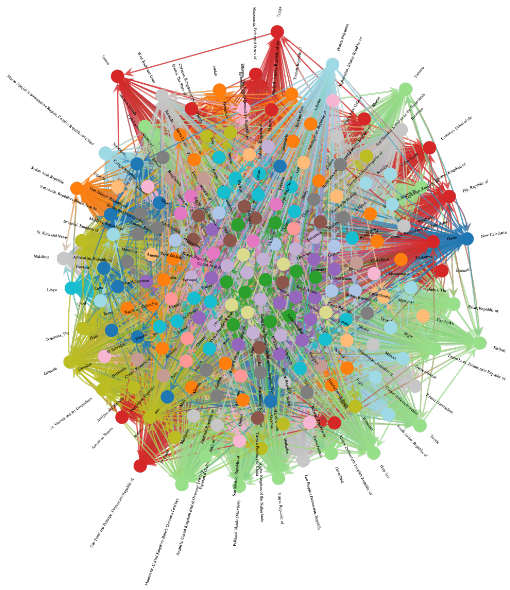
3.5 Alternative Model Structures

While the Flat Exponential model provided the most efficient fit according to MDL, the other tested models offer alternative perspectives on the network's structure. **Flat Log-Normal Model:** Our second place flat model was fitted with an assumed Log-Normal Distribution. This model featured 20 non-empty blocks (Table 3, detailed in Appendix B). In comparison to the Flat Exponential model we see this model's Block 22 overlaps significantly

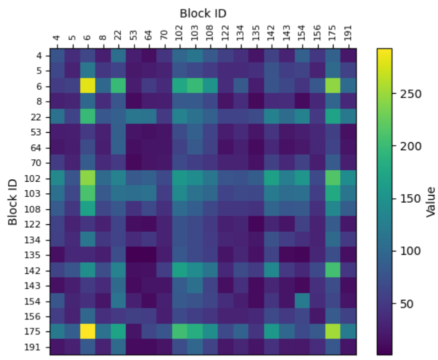
Table 6: Top & Bottom Inter-Block Export Flows (Flat Log-Normal Model)

Exporter	Importer	Ttl. Wght.	Edges
Top 5 by Total Export Value			
Block 175	Block 22	1110.04	169.00
Block 102	Block 22	1061.30	130.00
Block 22	Block 102	1054.25	130.00
Block 103	Block 103	994.73	110.00
Block 102	Block 102	963.44	156.00
Bottom 5 by Total Export Value (Non-Zero)			
Block 143	Block 154	0.26	17.00
Block 122	Block 53	0.23	23.00
Block 64	Block 154	0.17	9.00
Block 154	Block 8	0.11	9.00
Block 64	Block 135	0.06	1.00

Total Weight represents aggregate export value (millions USD).



(a) Flat Log-Normal SBM structure.



(b) Flat Log-Normal parameter heatmap.

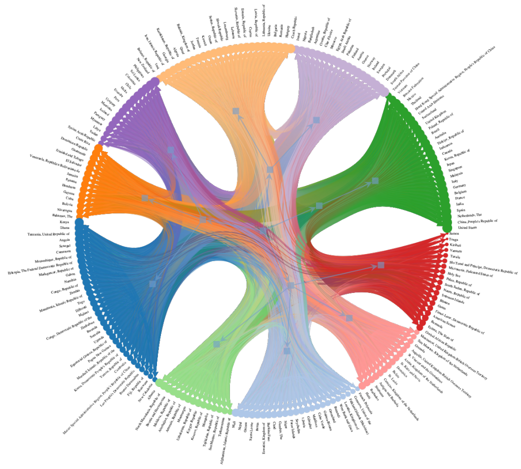
Figure 5: Flat Log-Normal model visualizations.

with the Flat Exponential's Block 49 (major global: US, China, Germany, etc.). However other groupings are different. If we look at the visualization (Figure 5a) we see the connections and partitions as differentiated colors. Looking at the trade heatmap (Figure 5b) we see the trade interactions between (horizontal and vertical stripes) and within the blocks (diagonal). The first stripe to look at is Block 6 (Mostly African Nations) with significant intra-block trade. Next is Block 22 (Major Global Traders). After that 102 (Czech Republic, Hungary, Romania, Israel, etc.) 103 (Asia-Pacific/Canada), 108 (Australia, Brazil, Mexico, Russia, etc.), 142 (Iran, Belarus, Uzbekistan), and 175 (Algeria, Oman, Lebanon) are worth consideration as larger trade groups as evidenced by the heat map. An interesting note is that intra-block trade in Block 6 (Mostly African Nations) and between Block 6 and Block 175 (Algeria, Oman, Lebanon) have the highest expected export amounts in this model. Table 6 details the top and bottom inter-block flows for this model.

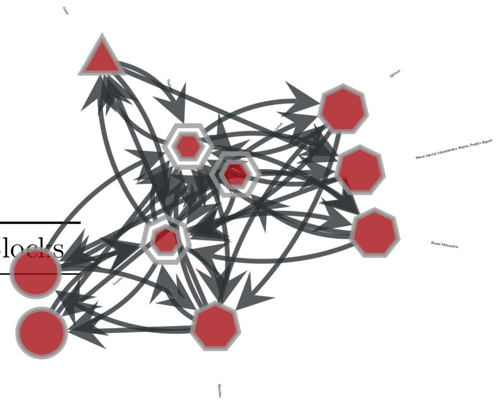
Table 7: Nested SBM Block Counts by Level and Model

Model	Lvl.	Total Blocks	Non-Empty Blocks
Nested Exp.	0	205	10
	1	8	4
	2	1	1
	3	1	1
	4	1	1
	5	1	1
	6	1	1
	7	1	1
	8	1	1
Nested Log.	0	191	13
	1	11	4
	2	1	1
	3	1	1
	4	1	1
	5	1	1
	6	1	1
	7	1	1
	8	1	1

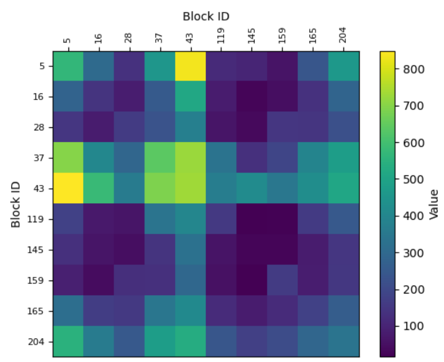
Nested Exponential Model: With the nested model we attempt a fit of the data while allowing hierarchical partitioning of blocks. A



(a) Full hierarchical structure inferred by the Nested Exponential WSBM.



(b) Nested Exponential Level 0 structure.



(c) Nested Exponential Level 0 parameter heatmap.

layer 0 block may be grouped with other layer 0 blocks within a level 1 consolidating block and on and on. This contrasts with the flat models one layer structure. We visualize this hierarchical structure in the Figure 6a. We first see the outer ring containing the names of all our 210 country nodes. We notice that they are grouped by color around the ring, this reflects the country memberships of the layer 0 blocks. If we inspect further towards the center, we can see blue squares with arrows pointing at them from a more central square. This represents the block at layer 0 and the membership of that block to a higher level block located closer to the center. A view of just these layer 0 blocks can be seen in Figure 6b showing 10 non-empty blocks at Level 0 (Table 7). If we look more deeply in the circle we see another layer of blue squares, layer 1, with 4 non-empty blocks (Table 7). Finally we see 1 non-empty block for layer 2 (Table 7).

Table 8: Top & Bottom Inter-Block Export Flows (Nested Exponential L0)

Exporter	Importer	Total Weight	Edges
Top 5 by Total Export Value			
Block 43	Block 43	6376.69	729.00
Block 37	Block 43	4012.43	687.00
Block 43	Block 37	3639.91	723.00
Block 43	Block 204	3482.14	531.00
Block 204	Block 43	3466.76	510.00
Bottom 5 by Total Export Value (Non-Zero)			
Block 145	Block 119	0.85	16.00
Block 159	Block 119	0.74	22.00
Block 159	Block 145	0.73	28.00
Block 145	Block 159	0.53	17.00
Block 119	Block 145	0.46	59.00

Total Weight represents aggregate export value (millions USD).

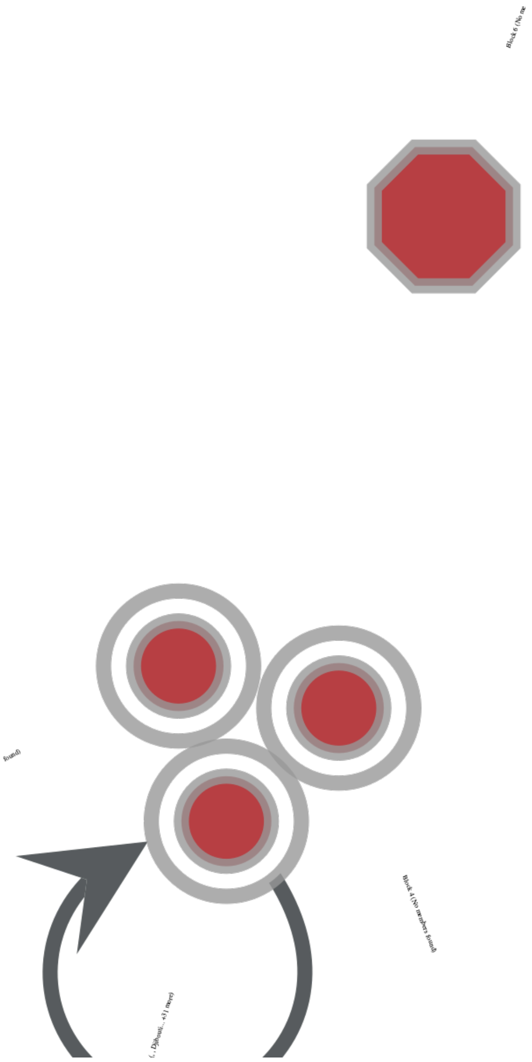
Similar to the Log-Normal visualization, the colored curved lines flowing between segments show the aggregate export flows, however these are hard to differentiate from the overlaid group colors and are most easily differentiated towards the middle. This visualization highlights the large economic core found in Block 43 (the large darker green segment between 1 and 3 o'clock) and its strong connections to other blocks, as well as the mainly African bloc in Block 5 (darker blue segment from 7 to 9 o'clock) and a mainly Latin American/Caribbean bloc in Block 28 (orange segment between 9 and 10 o'clock). The corresponding L0 parameter heatmap (Figure 6c) quantifies the average trade strength between these blocks. We observe strong intra-block trade within Block 43 (Total Weight 6376.69, Table 8) (Major Global/Asian/European economies). Significant inter-block trade is evident between Block 43 and Block 37 (Total Weights 4012.43 and 3639.91, Table 8) (grouping Eastern European, Middle east, and Central Asian countries) and Block 5 (mostly African). This model showed insights into the latent hierarchical structure level organization but its MDL (Table 2) at 72408.1542 was higher than the Flat Exponential model, implying that the added complexity of the hierarchy was not justified by the improvement in fit for this dataset.

Table 9: Top & Bottom Inter-Block Export Flows (Nested Log-Normal L0)

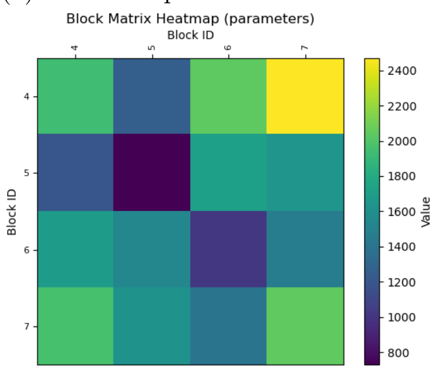
Exporter	Importer	Total Weight	Edges
Top 5 by Total Export Value			
Block 190	Block 190	1476.60	169.00
Block 190	Block 15	1207.41	140.00
Block 161	Block 190	1163.66	225.00
Block 161	Block 15	1152.72	179.00
Block 15	Block 190	1150.97	130.00
Bottom 5 by Total Export Value (Non-Zero)			
Block 161	Block 150	4.95	66.00
Block 68	Block 150	2.67	39.00
Block 150	Block 148	1.94	31.00
Block 150	Block 123	1.23	31.00
Block 123	Block 150	1.07	29.00

Total Weight represents aggregate export value (millions USD).

Further up the hierarchy, at Level 1 (Figure 7), the 10 Level 0 blocks are grouped into 4 larger super-blocks (visualized in Figure 7a). The Level 1 parameter heatmap (Figure 7b) shows the aggregated trade patterns between these four major groupings. We can observe the relative strength of trade flows connecting these higher-level structures, providing a



(a) Nested Exponential Level 1 structure.



(b) Nested Exponential Level 1 parameter heatmap.

Figure 7: Nested Exponential model visualizations (Level 1).

coarser view of the global trade network's organization as inferred by this nested model.

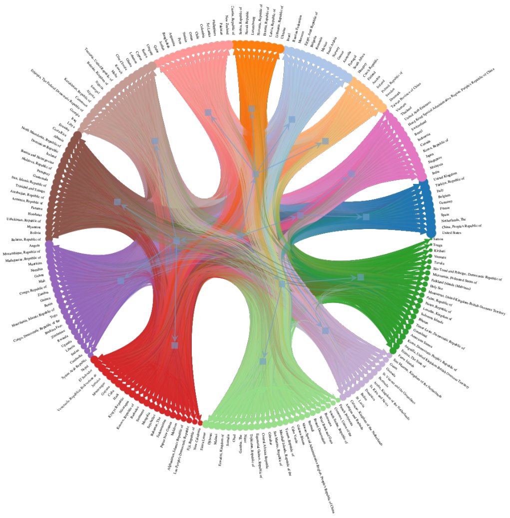
Nested Log-Normal Model: Like the Nested Exponential Model, this Log-Normal Model also sought to expose a latent hierarchy in the network. The Log-Normal model result in 13 non-empty blocks at the initial layer 0 (Table 7), vs 10 from the Exponential Model (visualized separately in Figure 8b, with memberships in Appendix D). The full hierarchy with squares showing the super-grouped block levels can be seen in Figure 8a.

Table 10: Top & Bottom Inter-Block Export Flows (Nested Log-Normal L1)

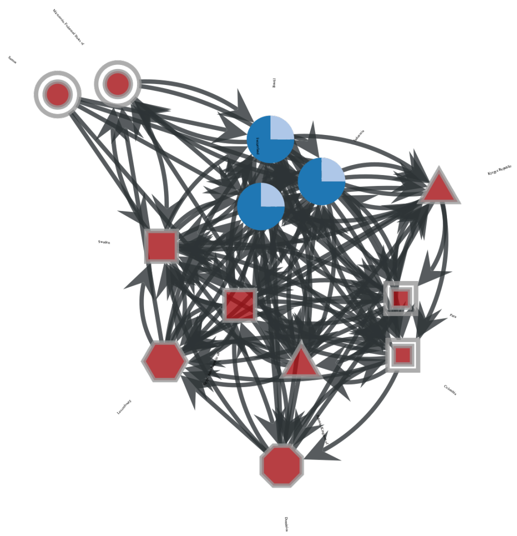
Exporter	Importer	Total Weight	Edges
Top 5 by Total Export Value			
Block 10	Block 10	9389.68	3861.00
Block 10	Block 0	7739.92	2962.00
Block 0	Block 10	7340.46	2902.00
Block 0	Block 0	5916.91	2164.00
Block 10	Block 9	4863.72	1956.00
Bottom 5 by Total Export Value (Non-Zero)			
Block 5	Block 0	2215.87	936.00
Block 0	Block 5	2158.39	923.00
Block 5	Block 9	1395.66	606.00
Block 9	Block 5	1385.67	625.00
Block 5	Block 5	714.35	352.00

Total Weight represents aggregate export value (millions USD).

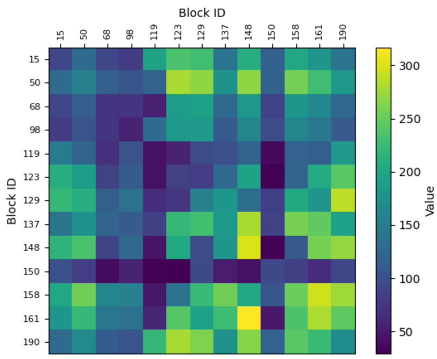
In this nested visualization we see that the major traders have been split into two groups in comparison with the Nested Exponential model. In blue between 2 and 3 o'clock, we see Block 15 (Major Globals, US, China, UK etc) and in magenta between 1 and 2 o'clock we see Block 190 (Japan, Singapore, Canada, Brazil, etc...) showing a more nuanced differentiation of these major trade countries. European and African nations are also partitioned differently in this model compared to the Nested Exponential model. Looking at the level 0 heatmap (Figure 8c), we see a different pattern than usual. In particular we don't see as many distinct horizontal stripes indicating significant inter-block trade for most blocks. The outstanding block in this case is Block 150 (Mostly Caribbean) which is lavender in Figure



(a) Full hierarchical structure inferred by the Nested Log-Normal WSBM.

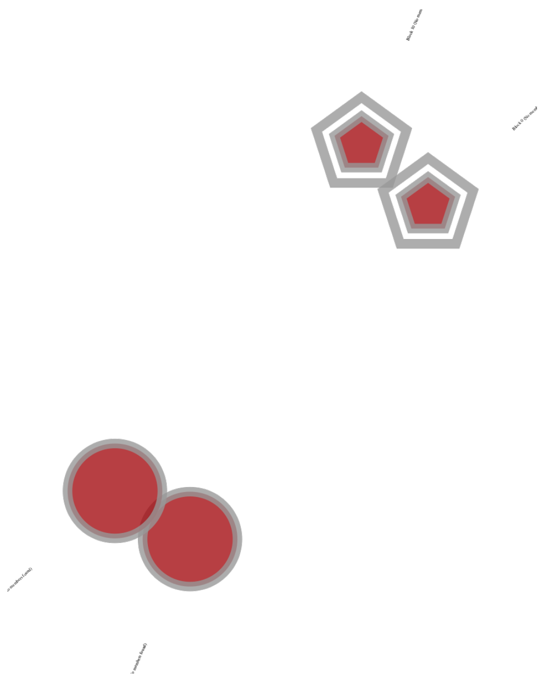


(b) Nested Log-Normal Level 0 structure.

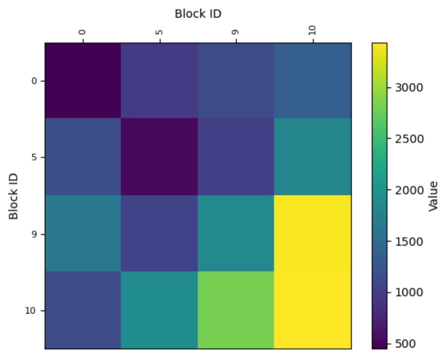


(c) Nested Log-Normal Level 0 parameter heatmap.

Figure 8: Nested Log-Normal model visualizations (Level 0).



(a) Nested Log-Normal Level 1 structure.



(b) Nested Log-Normal Level 1 parameter heatmap.

Figure 9: Nested Log-Normal model visualizations (Level 1).

8a. Block 150 shows low inter and intra block trade. The block with significant intra-block trade was Block 148 (African nations, cambodia, etc...). The blocks with significant inter-block trade were Blocks 15, 158, 161, and 190. The top and bottom inter-block flows for Level 0 are detailed in Table 9. At level 1, we have 4 non-empty blocks (Table 7). The most notable is Block 10, membership is referenced in Appendix D). It has outstanding intra-block trade with itself (Total Weight 9389.68, Table 10) and inter-block trade with Block 9 (Total Weight 4863.72, Table 10).

At Level 1 of the Nested Log-Normal hierarchy (Figure 9), the 13 Level 0 blocks (Table 7) merge into 4 Level 1 super-blocks (Table 7) (Figure 9a). The corresponding Level 1 parameter heatmap (Figure 9b) displays the expected trade volumes (see Table 10 for aggregate flows) between these four high-level clusters. This view aggregates the complex Level 0 interactions, showing the main channels of trade flow at a higher level of abstraction according to the Log-Normal assumption. However, this model had the highest MDL (Table 2, 79172.5098), suggesting it was the least efficient fit among the four configurations tested.

4 Discussion

We analyzed 2023 global export data using the python library graph-tool [5] and its WSBM to find the latent structure within. We approach the data with four model configurations, Flat vs Nested SBM type and Exponential vs Log-Normal covariate type. Using Minimum Description Length as our measure, "Flat Exponential" method worked best as the simplest and best fitting model. However exploring these other configurations like Nested and Log-Normal offered additional insights in the structure of the data, providing a view into the potential hierarchical nature and potential fit to a Log-Normal distribution.

By partitioning the countries into blocks, we were able to explore and derive insights about the grouped blocks, investigating block level metrics and inter-block and intra-block trade. A few consistencies were found. There were typically groups that contained members that acted as very large traders and acted as trading hubs connecting smaller groups with the global network. Blocks also showed regional patterns and size patterns, with regions being grouped together and countries of similar size and importance being grouped together. Many groupings also loosely reflected Trade partnerships and economic Blocs.

There are some considerations, we only used data from 2023 limiting the temporal view of this consistently evolving network. We also looked at total exports rather than having access to and exploring industry and province and state level data that would be more granular. Finally the methodology for approximating edge weights may have not perfectly represented the real world data, Exponential and Log-Normal distributions are simplified approximations of this distribution.

Future work may include expanding the study to more years, seeking more granular industry and state/province level data and integrating additional information about each country's relationships with each other, including trade groups and military alliances.

A Flat Exponential Block Memberships

Block 12: Croatia, Republic of, Estonia, Republic of, Latvia, Republic of, Lithuania, Republic of, Luxembourg, Romania, Serbia, Republic of, Slovak Republic, Slovenia, Republic of, Ukraine

Block 13: Argentina, Chile, Colombia, Ecuador, Guatemala, New Zealand, Peru, Uruguay

Block 18: Bangladesh, Egypt, Arab Republic of, Israel, Morocco, Pakistan, Saudi Arabia, South Africa

Block 21: Albania, Bosnia and Herzegovina, Iceland, Kosovo, Republic of, Moldova, Republic of, Montenegro, North Macedonia, Republic of

Block 31: Bahamas, The, Belize, Central African Republic, Curaçao, Kingdom of the Netherlands, Haiti, Macao Special Administrative Region, People's Republic of China, Suriname

Block 34: Bolivia, Brunei Darussalam, Fiji, Republic of, French Polynesia, Lao People's Democratic Republic, Papua New Guinea

Block 37: Congo, Democratic Republic of the, Djibouti, Ethiopia, The Federal Democratic Republic of, Kenya, Madagascar, Republic of, Mozambique, Republic of, Namibia, Sudan, Tanzania, United Republic of, Uganda, Zambia

Block 48: Armenia, Republic of, Faroe Islands, Greenland, Kyrgyz Republic, Mongolia, San Marino, Republic of, Tajikistan, Republic of, Turkmenistan

Block 49: Belgium, China, People's Republic of, France, Germany, India, Italy, Netherlands, The, Spain, Switzerland, Türkiye, Republic of, United Arab Emirates, United Kingdom, United States

Block 56: Algeria, Bahrain, Kingdom of, Jordan, Kuwait, Lebanon, Oman, Qatar, Tunisia

Block 58: Bhutan, Eritrea, The State of, Holy See, Kiribati, Micronesia, Federated States of, Nauru, Republic of, Samoa, Solomon Islands, Tonga, Vanuatu

Block 65: Botswana, Burundi, Comoros, Union of the, Eswatini, Kingdom of, Lesotho, Kingdom of, Malawi, Rwanda, Zimbabwe

Block 84: Angola, Burkina Faso, Cameroon, Congo, Republic of, Côte d'Ivoire, Gabon, Ghana, Korea, Democratic People's Republic of, Mauritania, Islamic Republic of, Mauritius, Nigeria, Senegal

Block 86: Chad, Gibraltar, Maldives, Marshall Islands, Republic of the, Nepal, New Caledonia, South Sudan, Republic of, West Bank and Gaza, Yemen, Republic of

Block 113: Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Barbados, Dominica, Grenada, Sint Maarten, Kingdom of the Netherlands, St. Lucia, St. Vincent and the Grenadines

Block 128: Cambodia, Costa Rica, Cuba, Dominican Republic, El Salvador, Guyana, Honduras, Jamaica, Nicaragua, Panama, Trinidad and Tobago, Venezuela, República Bolivariana de

Block 133: Bulgaria, Czech Republic, Finland, Hungary, Ireland, Portugal

Block 142: American Samoa, Anguilla, United Kingdom-British Overseas Territory, Bermuda, Guam, Montserrat, United Kingdom-British Overseas Territory, Palau, Republic of, St. Kitts and Nevis, Timor-Leste, Democratic Republic of, Tuvalu

Block 145: Cyprus, Libya, Malta, Myanmar, Paraguay, Philippines, Sri Lanka

Block 160: Australia, Brazil, Canada, Hong Kong Special Administrative Region, People's Republic of China, Indonesia, Japan, Korea, Republic of, Malaysia, Mexico, Singapore, Taiwan Province of China, Thailand, Vietnam

Block 184: Austria, Denmark, Greece, Norway, Poland, Republic of, Russian Federation, Sweden

Block 199: Cabo Verde, Falkland Islands (Malvinas), Gambia, The, Guinea-Bissau, Niger, Sierra Leone, São Tomé and Príncipe, Democratic Republic of

Block 201: Afghanistan, Islamic Republic of, Azerbaijan, Republic of, Belarus, Republic of, Georgia, Iran, Islamic Republic of, Iraq, Kazakhstan, Republic of, Syrian Arab Republic, Uzbekistan, Republic of

Block 205: Benin, Equatorial Guinea, Republic of, Guinea, Liberia, Mali, Seychelles, Somalia, Togo

B Flat Log-Normal Block Memberships

Block 4: Cuba, El Salvador, Guatemala, Guyana, Haiti, Jamaica, New Caledonia, Nicaragua, Panama, Suriname, Venezuela, República Bolivariana de

Block 5: Croatia, Republic of, Estonia, Republic of, Latvia, Republic of, Luxembourg, Serbia, Republic of, Ukraine

Block 6: Angola, Benin, Burkina Faso, Cameroon, Congo, Democratic Republic of the, Congo, Republic of, Djibouti, Gabon, Guinea, Liberia, Mali, Mauritania, Islamic Republic of, Namibia, Rwanda, Sierra Leone, Sudan, Syrian Arab Republic, Togo, Yemen, Republic of, Zambia, Zimbabwe

Block 8: Cambodia, Ethiopia, The Federal Democratic Republic of, Madagascar, Republic of, Mauritius, Mozambique, Republic of, Myanmar, Tanzania, United Republic of, Uganda

Block 22: Belgium, China, People's Republic of, France, Germany, Italy, Netherlands, The, Spain, Türkiye, Republic of, United Kingdom, United States

Block 53: Anguilla, United Kingdom-British Overseas Territory, Falkland Islands (Malvinas), Greenland, Holy See, Kiribati, Lesotho, Kingdom of, Montserrat, United Kingdom-British Overseas Territory, Nauru, Republic of, Palau, Republic of, Solomon Islands, Timor-Leste, Democratic Republic of, Tuvalu, Vanuatu

Block 64: American Samoa, Bhutan, Burundi, Comoros, Union of the, Eritrea, The State of, Fiji, Republic of, Guam, Korea, Democratic People's Republic of, Marshall Islands, Republic of the, Micronesia, Federated States of, Samoa, São Tomé and Príncipe, Democratic Republic of, Tonga

Block 70: Argentina, Chile, Colombia, Ecuador, Peru

Block 102: Bulgaria, Czech Republic, Finland, Greece, Hungary, Ireland, Israel, Lithuania, Republic of, Norway, Portugal, Romania, Slovak Republic, Slovenia, Republic of

Block 103: Canada, Hong Kong Special Administrative Region, People's Republic of China, India, Indonesia, Japan, Korea, Republic of, Malaysia, Singapore, Thailand, United Arab Emirates, Vietnam

Block 108: Australia, Brazil, Egypt, Arab Republic of, Mexico, Morocco, Russian Federation, Saudi Arabia, South Africa, Taiwan Province of China

Block 122: Bolivia, Costa Rica, Dominican Republic, Honduras, Paraguay, Trinidad and Tobago

Block 134: Bangladesh, New Zealand, Pakistan, Philippines, Qatar, Sri Lanka

Block 135: Afghanistan, Islamic Republic of, Faroe Islands, Gibraltar, Kyrgyz Republic, Montenegro, San Marino, Republic of, Turkmenistan

Block 142: Albania, Armenia, Republic of, Azerbaijan, Republic of, Belarus, Republic of, Bosnia and Herzegovina, Georgia, Iceland, Iran, Islamic Republic of, Kazakhstan, Republic of, Kosovo, Republic of, Moldova, Republic of, North Macedonia, Republic of, Uzbekistan, Republic of

Block 143: Botswana, Brunei Darussalam, Equatorial Guinea, Republic of, Lao People's Democratic Republic, Malawi, Maldives, Mongolia, Nepal, Papua New Guinea, Seychelles, West Bank and Gaza

Block 154: Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Bahamas, The, Barbados, Belize, Bermuda, Curaçao, Kingdom of the Netherlands, Dominica, Grenada, Sint Maarten, Kingdom of the Netherlands, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines

Block 156: Austria, Denmark, Poland, Republic of, Sweden, Switzerland

Block 175: Algeria, Bahrain, Kingdom of, Cyprus, Côte d'Ivoire, Ghana, Iraq, Jordan, Kenya, Kuwait, Lebanon, Libya, Malta, Nigeria, Oman, Senegal, Tunisia, Uruguay

Block 191: Cabo Verde, Central African Republic, Chad, Eswatini, Kingdom of, French Polynesia, Gambia, The, Guinea-Bissau, Macao Special Administrative Region, People's Republic of China, Niger, Somalia, South Sudan, Republic of, Tajikistan, Republic of

C Nested Exponential Block Memberships

C.1 Level 0

Block 5: Angola, Botswana, Brunei Darussalam, Cambodia, Cameroon, Congo, Democratic Republic of the, Congo, Republic of, Djibouti, Equatorial Guinea, Republic of, Ethiopia, The Federal Democratic Republic of, Fiji, Republic of, Gabon, Ghana, Kenya, Korea, Democratic People's Republic of, Lao People's Democratic Republic, Macao Special Administrative Region, People's Republic of China, Madagascar, Republic of, Malawi, Marshall Islands, Republic of the, Mauritania, Islamic Republic of, Mozambique, Republic of, Namibia, New Caledonia, Papua New Guinea, Rwanda, Senegal, Somalia, Tanzania, United Republic of, Togo, Uganda, Yemen, Republic of, Zambia, Zimbabwe

Block 16: Benin, Burkina Faso, Burundi, Cabo Verde, Chad, Comoros, Union of the, Eswatini, Kingdom of, Falkland Islands (Malvinas), Faroe Islands, French Polynesia, Gambia, The, Gibraltar, Greenland, Guinea, Guinea-Bissau, Lesotho, Kingdom of, Liberia, Maldives, Mali, Nepal, Niger, Seychelles, Sierra Leone, West Bank and Gaza

Block 28: Bahamas, The, Bolivia, Costa Rica, Cuba, Dominican Republic, El Salvador, Guatemala, Guyana, Honduras, Jamaica, Nicaragua, Panama, Trinidad and Tobago, Venezuela, República Bolivariana de

Block 37: Algeria, Bahrain, Kingdom of, Belarus, Republic of, Bulgaria, Cyprus, Czech Republic, Estonia, Republic of, Georgia, Hungary, Iran, Islamic Republic of, Iraq, Jordan, Kazakhstan, Republic of, Kuwait, Latvia, Republic of, Lebanon, Lithuania, Republic of, Luxembourg, Oman, Qatar, Romania, Serbia, Republic of, Slovak Republic, Slovenia, Republic of, Tunisia, Ukraine

Block 43: Australia, Belgium, Brazil, Canada, China, People's Republic of, France, Germany, Hong Kong Special Administrative Region, People's Republic of China, India, Indonesia, Italy, Japan, Korea, Republic of, Malaysia, Mexico, Netherlands, The, Poland, Republic of, Russian Federation, Singapore, Spain, Switzerland, Taiwan Province of China, Thailand, Türkiye, Republic of, United Arab Emirates, United Kingdom, United States, Vietnam

Block 119: Afghanistan, Islamic Republic of, Albania, Armenia, Republic of, Azerbaijan, Republic of, Bosnia and Herzegovina, Kosovo, Republic of, Kyrgyz Republic, Moldova, Republic of, Mongolia, Montenegro, North Macedonia, Republic of, San Marino, Republic of, Tajikistan, Republic of, Turkmenistan, Uzbekistan, Republic of

Block 145: American Samoa, Bermuda, Bhutan, Central African Republic, Eritrea, The State of, Guam, Holy See, Kiribati, Micronesia, Federated States of, Nauru, Republic of, Palau, Republic of, Samoa, Solomon Islands, South Sudan, Republic of, São Tomé and Príncipe, Democratic Republic of, Timor-Leste, Democratic Republic of, Tonga, Tuvalu, Vanuatu

Block 159: Anguilla, United Kingdom-British Overseas Territory, Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Barbados, Belize, Curaçao, Kingdom of the Netherlands, Dominica, Grenada, Haiti, Montserrat, United Kingdom-British Overseas Territory, Sint Maarten, Kingdom of the Netherlands, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, Suriname

Block 165: Chile, Colombia, Ecuador, Iceland, Libya, Malta, Mauritius, Myanmar, New Zealand, Paraguay, Peru, Philippines, Sri Lanka, Sudan, Syrian Arab Republic, Uruguay

Block 204: Argentina, Austria, Bangladesh, Croatia, Republic of, Côte d'Ivoire, Denmark, Egypt, Arab Republic of, Finland, Greece, Ireland, Israel, Morocco, Nigeria, Norway, Pakistan, Portugal, Saudi Arabia, South Africa, Sweden

C.2 Level 1

Block 4: Angola, Azerbaijan, Republic of, Belarus, Republic of, Bermuda, Botswana, Bulgaria, Cabo Verde, Central African Republic, Congo, Democratic Republic of the, Costa Rica, Croatia, Republic of, Cuba, Denmark, Dominican Republic, Egypt, Arab Republic of, Gabon, Gibraltar, Guam, Guyana, Holy See, Hungary, Iraq, Jamaica, Jordan, Kenya, Korea, Republic of, Kosovo, Republic of, Lebanon, Luxembourg, Maldives, Mali, Malta, Montenegro, Panama, Papua New Guinea, Philippines, Qatar, Saudi Arabia, Sierra Leone, Sudan, Syrian Arab Republic, Turkmenistan, Türkiye, Republic of, Uganda, Ukraine

Block 5: Afghanistan, Islamic Republic of, American Samoa, Argentina, Armenia, Republic of, Bhutan, Congo, Republic of, Eritrea, The State of, Finland, Iceland, Latvia, Republic of, Madagascar, Republic of, Marshall Islands, Republic of the, Moldova, Republic of, Morocco, Myanmar, Nauru, Republic of, Netherlands, The, Pakistan, Paraguay, Romania, Russian Federation, San Marino, Republic of, South Africa, St. Lucia, Suriname, Thailand, United Kingdom, West Bank and Gaza

Block 6: Albania, Anguilla, United Kingdom-British Overseas Territory, Bahamas, The, Bahrain, Kingdom of, Bangladesh, Belgium, Benin, Bosnia and Herzegovina, Brunei Darussalam, Burkina Faso, Burundi, Cambodia, Cameroon, Chad, China, People's Republic of, Curaçao, Kingdom of the Netherlands, Djibouti, El Salvador, Eswatini, Kingdom of, Ethiopia, The Federal Democratic Republic of, Falkland Islands (Malvinas), Faroe Islands, Fiji, Republic of, France, Gambia, The, Georgia, Ghana, Greece, Greenland, Grenada, Guinea-Bissau, Haiti, Honduras, Indonesia, Iran, Islamic Republic of, Ireland, Israel, Italy, Korea, Democratic People's Republic of, Kuwait, Lao People's Democratic Republic, Lesotho, Kingdom of, Liberia, Libya, Lithuania, Republic of, Mexico, Mongolia, Nepal, New Caledonia, Nicaragua, Niger, North Macedonia, Republic of, Oman, Peru, Portugal, Samoa, Seychelles, Singapore, Slovak Republic, Somalia, St. Kitts and Nevis, St. Vincent and the Grenadines, Sweden, Taiwan Province of China, Tanzania, United Republic of, Timor-Leste, Democratic Republic of, Tonga, Tuvalu, Uruguay, Uzbekistan, Republic of, Vanuatu, Yemen, Republic of, Zambia

Block 7: Algeria, Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Australia, Austria, Barbados, Belize, Bolivia, Brazil, Canada, Chile, Colombia, Comoros, Union of the, Cyprus, Czech Republic, Côte d'Ivoire, Dominica, Ecuador, Equatorial Guinea, Republic of, Estonia, Republic of, French Polynesia, Germany, Guatemala, Guinea, Hong Kong Special Administrative Region, People's Republic of China, India, Japan, Kazakhstan, Republic of, Kiribati, Kyrgyz Republic, Macao Special Administrative Region, People's Republic of China, Malawi, Malaysia, Mauritania, Islamic Republic of, Mauritius, Micronesia, Federated States of, Montserrat, United Kingdom-British Overseas Territory, Mozambique, Republic of, Namibia, New Zealand, Nigeria, Norway, Palau, Republic of, Poland, Republic of, Rwanda, Senegal, Serbia, Republic of, Sint Maarten, Kingdom of the Netherlands, Slovenia, Republic of, Solomon Islands, South Sudan, Republic of, Spain, Sri Lanka, Switzerland, São Tomé and Príncipe, Democratic Republic of, Tajikistan, Republic of, Togo, Trinidad and Tobago, Tunisia, United Arab Emirates, United States, Venezuela, República Bolivariana de, Vietnam, Zimbabwe

C.3 Levels 2-8

Note: Block memberships for Levels 2 through 8 contain all 210 countries in a single block.

D Nested Log-Normal Block Memberships

D.1 Level 0

Block 15: Belgium, China, People's Republic of, France, Germany, Italy, Netherlands, The, Spain, Türkiye, Republic of, United Kingdom, United States

Block 50: Australia, Bulgaria, Egypt, Arab Republic of, Greece, Israel, Mexico, Morocco, Norway, Portugal, Romania, Russian Federation, Saudi Arabia, South Africa

Block 68: Croatia, Republic of, Estonia, Republic of, Latvia, Republic of, Lithuania, Republic of, Luxembourg, Serbia, Republic of, Slovak Republic, Slovenia, Republic of, Ukraine

Block 98: Austria, Czech Republic, Denmark, Finland, Hungary, Ireland, Poland, Republic of, Sweden

Block 119: American Samoa, Anguilla, United Kingdom-British Overseas Territory, Bhutan, Eritrea, The State of, Falkland Islands (Malvinas), Faroe Islands, Greenland, Holy See, Kiribati, Korea, Democratic People's Republic of, Lesotho, Kingdom of, Micronesia, Federated States of, Montserrat, United Kingdom-British Overseas Territory, Nauru, Republic of, Palau, Republic of, Samoa, Solomon Islands, São Tomé and Príncipe, Democratic Republic of, Timor-Leste, Democratic Republic of, Tonga, Tuvalu, Vanuatu

Block 123: Botswana, Brunei Darussalam, Burundi, Cabo Verde, Central African Republic, Chad, Comoros, Union of the, Djibouti, Equatorial Guinea, Republic of, Eswatini, Kingdom of, French Polynesia, Gambia, The, Gibraltar, Guinea-Bissau, Macao Special Administrative Region, People's Republic of China, Malawi, Marshall Islands, Republic of the, Niger, San Marino, Republic of, Sierra Leone, Somalia, South Sudan, Republic of, Tajikistan, Republic of, West Bank and Gaza, Yemen, Republic of

Block 129: Afghanistan, Islamic Republic of, Bahamas, The, Barbados, Cuba, El Salvador, Fiji, Republic of, Guyana, Haiti, Jamaica, Kosovo, Republic of, Kyrgyz Republic, Lao People's Democratic Republic, Maldives, Mongolia, Montenegro, Nepal, New Caledonia, Nicaragua, Papua New Guinea, Seychelles, Suriname, Turkmenistan, Venezuela, República Bolivariana de

Block 137: Argentina, Bangladesh, Chile, Colombia, Jordan, New Zealand, Oman, Pakistan, Peru, Philippines, Qatar, Sri Lanka, Tunisia, Uruguay

Block 148: Angola, Benin, Burkina Faso, Cambodia, Congo, Democratic Republic of the, Congo, Republic of, Gabon, Guinea, Liberia, Madagascar, Republic of, Mali, Mauritania, Islamic Republic of, Mauritius, Mozambique, Republic of, Namibia, Rwanda, Sudan, Syrian Arab Republic, Togo, Uganda, Zambia, Zimbabwe

Block 150: Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Belize, Bermuda, Curaçao, Kingdom of the Netherlands, Dominica, Grenada, Guam, Sint Maarten, Kingdom of the Netherlands, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines

Block 158: Albania, Armenia, Republic of, Azerbaijan, Republic of, Belarus, Republic of, Bolivia, Bosnia and Herzegovina, Costa Rica, Dominican Republic, Ecuador, Guatemala, Honduras, Iceland, Iran, Islamic Republic of, Moldova, Republic of, Myanmar, North Macedonia, Republic of, Panama, Paraguay, Trinidad and Tobago, Uzbekistan, Republic of

Block 161: Algeria, Bahrain, Kingdom of, Cameroon, Cyprus, Côte d'Ivoire, Ethiopia, The Federal Democratic Republic of, Georgia, Ghana, Iraq, Kazakhstan, Republic of, Kenya, Kuwait, Lebanon, Libya, Malta, Nigeria, Senegal, Tanzania, United Republic of

Block 190: Brazil, Canada, Hong Kong Special Administrative Region, People's Republic of China, India, Indonesia, Japan, Korea, Republic of, Malaysia, Singapore, Switzerland, Taiwan Province of China, Thailand, United Arab Emirates, Vietnam

D.2 Level 1

Block 0: Algeria, Angola, Antigua and Barbuda, Argentina, Australia, Bahamas, The, Belgium, Benin, Bhutan, Bosnia and Herzegovina, Chad, Comoros, Union of the, Congo, Democratic Republic of the, Cuba, Cyprus, Czech Republic, Côte d'Ivoire, Dominica, Ecuador, Estonia, Republic of, Ethiopia, The Federal Democratic Republic of, Falkland Islands (Malvinas), Finland, French Polynesia, Gabon, Guam, Guyana, Ireland, Japan, Kazakhstan, Republic of, Korea, Democratic People's Republic of, Kosovo, Republic of, Lao People's Democratic Republic, Madagascar, Republic of, Malta, Mauritania, Islamic Republic of, Mauritius, Mexico, Mongolia, Netherlands, The, Norway, Poland, Republic of, Rwanda, Serbia, Republic of, Sierra Leone, Somalia, South Africa, South Sudan, Republic of, Sweden, Switzerland, Tajikistan, Republic of, Tonga, Tunisia, Türkiye, Republic of, Uganda, United Arab Emirates, United Kingdom, Vanuatu, Zimbabwe

Block 5: Afghanistan, Islamic Republic of, Albania, Anguilla, United Kingdom-British Overseas Territory, Belize, Brazil, China, People's Republic of, Costa Rica, Egypt, Arab Republic of, Georgia, Ghana, Honduras, Kenya, Latvia, Republic of, Lebanon, Maldives, Moldova, Republic of, Namibia, New Zealand, St. Kitts and Nevis, St. Lucia, Sudan, Tanzania, United Republic of, Venezuela, República Bolivariana de, Zambia

Block 9: Armenia, Republic of, Azerbaijan, Republic of, Bangladesh, Belarus, Republic of, Bolivia, Burundi, Cabo Verde, Cameroon, Colombia, Denmark, Faroe Islands, Guinea, Guinea-Bissau, Holy See, Hungary, Indonesia, Iran, Islamic Republic of, Iraq, Israel, Jamaica, Kiribati, Kyrgyz Republic, Liberia, Micronesia, Federated States of, Montserrat, United Kingdom-British Overseas Territory, Morocco, Myanmar, Nicaragua, Niger, Oman, Panama, Portugal, Qatar, Russian Federation, San Marino, Republic of, Sint Maarten, Kingdom of the Netherlands, Slovenia, Republic of, St. Vincent and the Grenadines, Syrian Arab Republic, Tuvalu, Ukraine, United States, Vietnam, Yemen, Republic of

Block 10: American Samoa, Aruba, Kingdom of the Netherlands, Austria, Bahrain, Kingdom of, Barbados, Bermuda, Botswana, Brunei Darussalam, Bulgaria, Burkina Faso, Cambodia, Canada, Central African Republic, Chile, Congo, Republic of, Croatia, Republic of, Curaçao, Kingdom of the Netherlands, Djibouti, Dominican Republic, El Salvador, Equatorial Guinea, Republic of, Eritrea, The State of, Eswatini, Kingdom of, Fiji, Republic of, France, Gambia, The, Germany, Gibraltar, Greece, Greenland, Grenada, Guatemala, Haiti, Hong Kong Special Administrative Region, People's Republic of China, Iceland, India, Italy, Jordan, Korea, Republic of, Kuwait, Lesotho, Kingdom of, Libya, Lithuania, Republic of, Luxembourg, Macao Special Administrative Region, People's Republic of China, Malawi, Malaysia, Mali, Marshall Islands, Republic of the, Montenegro, Mozambique, Republic of, Nauru, Republic of, Nepal, New Caledonia, Nigeria, North Macedonia, Republic of, Pakistan, Palau, Republic of, Papua New Guinea, Paraguay, Peru, Philippines, Romania, Samoa, Saudi Arabia, Senegal, Seychelles, Singapore, Slovak Republic, Solomon Islands, Spain, Sri Lanka, Suriname, São Tomé and Príncipe, Democratic Republic of, Taiwan Province of China, Thailand, Timor-Leste, Democratic Republic of, Togo, Trinidad and Tobago, Turkmenistan, Uruguay, Uzbekistan, Republic of, West Bank and Gaza

D.3 Levels 2-8

Note: Block memberships for Levels 2 through 8 contain all 210 countries in a single block.

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